

# When Is the Mover-Design Event Study Coefficient a Place-Effect Share?

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## Abstract

Mover-design event studies are a leading approach to separating place effects from sorting. The coefficient is often interpreted as the share of cross-sectional variation in location means due to places. I show the coefficient depends on how movers connect locations: two economies can be identical in place effects, sorting, and location means, yet deliver different mover coefficients. Only under *directional isotropy*, a testable condition, does the coefficient permit a cross-sectional reading: it averages the variance share of place effects and the share that equalizing them removes. An illustration in Dutch employer–employee wage data rejects the condition.

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## I Introduction

Outcomes differ greatly across firms, regions, hospitals, and schools, but separating genuine place effects from sorting of people into places is difficult. The mover-design event study is a leading approach to this problem (Finkelstein et al., 2016). By tracking individuals who change locations, this design relates the change in a mover’s outcome to the difference in average outcomes between their destination and origin. The resulting coefficient captures how much of the gap between locations passes to the individual upon arrival. This estimate is often interpreted as the contribution of place effects to the variation in mean outcomes across locations. The design is widely used in health, public economics, political economy, and labor economics.<sup>1</sup>

This paper provides the first formal link between the mover-design coefficient and cross-sectional variance components. I study the design in the standard Abowd–Kramarz–Margolis (AKM) setting (Abowd et al., 1999)—outcomes are additively separable in individual and place effects—with large locations and a stationary cross-section, as in most applications,<sup>2</sup> while remaining agnostic about the underlying sorting and mobility process. The main result is that the mover-design coefficient is a transition-weighted projection: its moments are determined by where movers come from and where they go, not by the cross-sectional distribution of places. Two economies can share the same AKM decomposition—the same place-effect dispersion, individual-effect dispersion, sorting, and location-mean dispersion—and still produce different mover coefficients. The cross-section describes the map. The mover coefficient depends on the particular routes taken through it.

In a mover-design event study, the regressor is the change in location-level mean outcomes. A location mean combines the place effect with *composition*: the average individual effect of the people who sort into that location. Composition has two components. One is correlated with the place effect, as when higher-paying firms employ better workers. The other, a residual, is orthogonal to the place effect, as when two firms with the same pay premium hire different types of workers because they differ in amenities, industry, hours, or career structure. How much a typical move changes each part depends on who moves where, and need not mirror how the two parts vary across locations. In employer–employee settings, a worker who climbs a job ladder and moves to a higher-paying firm with similar amenities mainly moves along the place-effect dimension. A worker who switches careers and moves to a similar-paying firm with very different coworkers mainly moves along the residual-composition dimension. These moves contribute differently to the mover coefficient, even if the cross-sectional distribution of firms is unchanged. In simulations, I construct

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<sup>1</sup>Applications include studies of geographic variation in health care (Finkelstein et al., 2016; Godøy and Huitfeldt, 2020; Salm and Wübker, 2020; Zeltzer et al., 2021; Luan et al., 2025), physician and organizational behavior (Molitor, 2018; Huang and Ullrich, 2024; Doyle and Staiger, 2025; Agha et al., 2023; Encinosa and Dor, 2025; Badinski et al., 2023), health behaviors and outcomes (Hinnsaar and Liu, 2022; Hut, 2020; Finkelstein et al., 2025; Ding, 2026; Chyn and Shenhav, 2025), insurance and program choice (Cabral et al., 2025), employer and research institution effects (Ahammer et al., 2024; Chandra and Xu, 2025), political behavior (Cantoni and Pons, 2022), and carbon emissions (Lyubich, 2025).

<sup>2</sup>Throughout, the *cross-section* is the snapshot the researcher observes: each individual’s fixed effect, each location’s place effect, the composition of individuals within it, and hence its mean outcome. *Stationarity* means this snapshot is stable across periods; substantial worker churning can coexist with a nearly fixed cross-section, since composition moves only through the net imbalance of gross inflows and outflows.

economies with identical place effects, sorting, location means, and pooled AKM decompositions, and vary only the network of realized moves. The mover-design event study coefficient varies substantially.

The mover-design coefficient recovers the cross-section only when mobility is *directionally isotropic*: movers must traverse the place-effect dimension and the residual-composition dimension with the same intensity as those dimensions vary in the cross-section—movers must sample the map in the proportions the map varies. Under this condition, the coefficient equals the cross-sectional regression of place effects on location means, and this object has a sharp interpretation. It is the average of two shares of the cross-sectional variance of location means—the share attributable to place effects, and the share that would disappear if place effects were equalized. Under isotropy, two limiting configurations recover familiar readings. At one extreme, if composition is uncorrelated with place effects, the coefficient is the variance share of place effects. This case corresponds to sorting on factors unrelated to the outcome under study: in a setting like that of [Finkelstein et al. \(2016\)](#), patients who settled where they do for family, retirement, or housing—rather than for health care—would generate composition orthogonal to the place effect. At the other extreme, if composition is *affine*—linear and residual-free—in the place effect, the coefficient is the standard-deviation share. This case corresponds to matching on the outcome dimension itself, as when workers value jobs only through pay and a location’s average worker type rises linearly, with no residual component, in its premium. Under affine sorting, a coefficient of 0.20—often described as “places explain twenty percent of the variation”—corresponds to a variance contribution of four percent.<sup>3</sup> The affine reading is special in a second way: it is robust to *any* mobility pattern.

Whether mobility satisfies isotropy is an empirical question. The condition restricts how cross-sectional and mobility moments relate in the data. I provide a test for isotropy—a cross-fitted diagnostic with an individual-level bootstrap—and illustrate it on the mobility of one-time movers across large firms in Dutch matched employer–employee data. Sorting on the firm wage premium is statistically undetectable in the cross-section and negative along moves; movers, however, traverse the residual-composition dimension about a third more intensively than its cross-sectional dispersion implies, and isotropy is rejected. The mover-design coefficient therefore cannot be mapped to a share of the variation of location means.<sup>4</sup>

The scaled mover event study originates with [Bronnenberg et al. \(2012\)](#), where the share of the origin–destination gap that closes upon moving is a structural constant; the jump then measures a cross-market share by construction, and the interpretive question is settled by the model. [Finkelstein et al. \(2016\)](#) bring the design to the two-way fixed-effects setting, and it is there that the cross-sectional reading becomes a focal point. They read the jump as a weighted average of pair-level place-effect shares and bridge to the cross-sectional variation share through

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<sup>3</sup>E.g., “[l]ocation explains 37 percent of the cross-state variation in turnout” ([Cantoni and Pons, 2022](#), abstract); similarly, [Molitor \(2018\)](#) reads a mover coefficient of 0.6–0.8 as environment factors explaining “between 60–80 percent of regional disparities” in physician behavior.

<sup>4</sup>Throughout, “cross-sectional” moments are weighted by mover incidence: the cross-section taken over the locations movers actually visit. Section [IV](#) makes the weighting explicit.

the observed linearity of movers' outcome changes in the origin–destination gap. I show that this bridge is not generic—it holds only under directional isotropy—and in its place I derive an exact mapping from the coefficient to the cross-sectional variance components.<sup>5</sup> [Hull \(2018\)](#) asks what the mover regression recovers *causally*, showing it aggregates difference-in-differences comparisons with convex weights under a parallel-trends assumption. [Card et al. \(2025\)](#) confront a similar contamination—a move across regions also changes the firm, so firm sorting biases the regional (unscaled) movers design—which they resolve by estimating a model with worker and firm effects and aggregating to the commuting-zone level.<sup>6</sup> This paper instead asks what the mover-design event study coefficient estimates, with no structure imposed, and the characterization is complete under stationarity: the estimand depends only on the cross-sectional sorting geometry and the undirected mobility graph, and every admissible combination of the two is generated by some assignment process in the AKM class. Whether the cross-sectional reading is valid is therefore a property of the reduced form alone—no assignment model is needed to test for it.<sup>7</sup>

## II Model and Setup

Consider a balanced panel of individuals  $i = 1, \dots, N$  observed in two periods,  $t \in \{1, 2\}$ , each residing in one of  $J$  locations  $j \in \mathcal{J}$ ;  $j(i, t)$  denotes  $i$ 's location at  $t$ .<sup>8</sup> I study an asymptotic regime with fixed  $J$  and large location sizes  $n_j \rightarrow \infty$ , which matches the empirical setting of mover-design event studies and separates the mobility mechanisms studied here from limited-mobility bias.

Outcomes follow the standard AKM,

$$Y_{it} = \alpha_i + \psi_{j(i,t)} + \varepsilon_{it}, \quad t \in \{1, 2\}, \quad (1)$$

with time-invariant individual effects  $\alpha_i$ , time-invariant location effects  $\psi_j$ , and idiosyncratic shocks satisfying the exogenous-mobility restriction  $\mathbb{E}[\varepsilon_{it} \mid j(i, k) = j] = 0$  for all  $(k, t)$  and  $j$ . This rules out mobility driven by realized or anticipated idiosyncratic shocks but places *no* restriction on the joint distribution of  $(\alpha_i, \psi_{j(i,t)})$ . Individuals may sort across locations without restriction. Error vectors are independent across individuals with bounded second moments; within an individual,  $\varepsilon_{i1}$  and

<sup>5</sup>For example, if the place-effect and composition changes are jointly elliptically distributed across moves, the average outcome change is exactly linear in the origin–destination gap whatever the sorting geometry: the geometry fixes the slope of the mover plot. A straight plot is thus consistent with every geometry, and cannot on its own separate a valid cross-sectional reading from an invalid one.

<sup>6</sup>In labor economics, mover designs often serve a different role. In their unscaled, co-worker-quartile form, they assess exogenous mobility and additive separability ([Card et al., 2013, 2016](#)), while variance shares come from the AKM decomposition ([Abowd et al., 1999](#)).

<sup>7</sup>The object of interest in this paper is also distinct from limited-mobility bias ([Andrews et al., 2008](#); [Kline et al., 2020](#); [Bonhomme et al., 2023](#))—noise in estimated place effects when movers are sparse—which vanishes in the large-location regime of this paper.

<sup>8</sup>Supplemental Appendix S1 shows that the canonical multi-period mover-design event study of [Finkelstein et al. \(2016\)](#) is asymptotically equivalent to this two-period representation: pre-event coefficients converge to zero and all post-event coefficients converge to the two-period first-difference estimand. When the AKM specification includes time-varying controls, every object below is defined on the control-adjusted outcome  $Y_{it} - X'_{it}\gamma$ , and all recovered shares are shares of *control-adjusted* variation; see the remark there.

$\varepsilon_{i2}$  may be arbitrarily correlated.

Write  $o(i) := j(i, 1)$  and  $d(i) := j(i, 2)$  for origin and destination,  $M := \{i : o(i) \neq d(i)\}$  for movers,  $S$  for stayers, and  $\Delta\psi_i := \psi_{d(i)} - \psi_{o(i)}$ . Movers and stayers share the outcome process (1) but may differ in their effect distributions and sorting.

**Assumption 1** (Stationarity). Location sizes and mover shares are constant across periods,  $n_{j,1} = n_{j,2} \equiv n_j$  and  $w_{j,1} = w_{j,2} \equiv w_j$ , and for each location  $j$  the distribution of  $\alpha_i$  among movers leaving  $j$  equals the distribution among movers arriving at  $j$ .

Assumption 1 says that locations are in a demographic steady state: each keeps its size, its mover share, and the type profile of its mover population stable across periods—the people arriving resemble the people leaving. What it does *not* restrict is the object the design exploits, who moves where: origins and destinations may pair arbitrarily, and movers with different  $\alpha_i$  may take different routes, provided each location’s margins balance. Supplemental Appendix S2 characterizes the implied balance conditions on mover flows.

Stationarity is a premise, not a necessity. Everything that follows—the estimand, the recovery characterization, and the isotropy test—survives without it, read against the average of the two periods’ cross-sections, with the location means in the regressor averaged over the window as in applied implementations; the one exception is the reduction of mobility to an undirected flow, which uses the mass balance that stationarity supplies (Supplemental Appendix S3). Stationarity is also the *least* favorable case for the thesis here: relaxing it only enlarges the admissible flows—drifting configurations now qualify—so the spread of coefficients consistent with a given cross-section can only grow. The assumption is nonetheless the natural benchmark for the settings where mover designs are applied. Realized mobility can be thought of as a circulation that preserves the cross-section plus a residual that carries its drift, and the circulation dominates. A location’s composition moves only through the net of offsetting inflows and outflows—second-order where gross flows dwarf net flows. For example, employee–employer wage-data cross-sections drift over decadal horizons, workers churn annually (cf. Card et al., 2013). Assumption 1 therefore holds fixed what remains approximately stable in the data. It treats circulation as the object of the theory and interprets the two period-specific cross-sections as realizations of a common cross-section, thereby rendering the cross-sectional benchmarks well defined.

**Sorting and composition.** I summarize sorting by projecting individual effects onto the location-effect dimension. For each group  $g \in \{M, S\}$ ,

$$\alpha_i = m_g(\psi_{j(i,t)}) + v_{j(i,t),g} + e_{i,t,g}, \quad (2)$$

where  $m_g(\psi) := \mathbb{E}[\alpha_i \mid \psi_{j(i,t)} = \psi, i \in g]$  captures sorting that is systematic in  $\psi_j$ ,  $v_{j,g}$  (mean zero given  $\psi_j$ ) captures persistent composition differences across locations with the same  $\psi_j$ , and  $e_{i,t,g}$  is the within-location individual residual. The function  $m_g$  captures sorting on the location effect

itself and on attributes correlated with it—amenities, hours, job security, geography, career paths. The residual  $v_{j,g}$  allows two locations with the same effect to attract different types of individuals.

Averaging (2) within location  $j$  in the large-location regime gives the location's composition,

$$\bar{\alpha}_j = H_j(\psi_j) + v_j, \quad H_j(\psi_j) := w_j m_M(\psi_j) + (1 - w_j) m_S(\psi_j), \quad v_j := w_j v_{j,M} + (1 - w_j) v_{j,S}, \quad (3)$$

time-invariant by Assumption 1.  $H_j(\psi_j)$  is the part of composition predictable from the location-effect dimension;  $v_j$  is the remaining *residual composition*, orthogonal to that dimension in the cross-section.

### III The Mover-Design Estimand

Among movers, the researcher regresses the individual outcome change on the leave-one-out change in location mean outcomes:

$$\Delta Y_i = \theta \delta_i^{(-i)} + u_i, \quad i \in M, \quad \delta_i^{(-i)} := \bar{Y}_{d(i),2}^{(-i)} - \bar{Y}_{o(i),1}^{(-i)}, \quad (4)$$

where  $\bar{Y}_{j,t}^{(-i)}$  is the leave-one-out mean outcome in cell  $(j, t)$ . Under (1),  $\Delta Y_i = \Delta \psi_i + \Delta \varepsilon_i$ , and exogenous mobility makes  $\Delta \varepsilon_i$  orthogonal to the regressor. The regressor, however, is not the change in location effects; in the large-location limit,

$$\delta_i^{(-i)} = \Delta \psi_i + \Delta \bar{\alpha}_i, \quad \Delta \bar{\alpha}_i := \bar{\alpha}_{d(i)} - \bar{\alpha}_{o(i)}.$$

The dependent variable changes only because the mover changes location effects; the regressor changes because the mover changes both location effects and location composition. The large-location estimand is therefore

$$\theta_\infty = \frac{\text{Var}_M(\Delta \psi_i) + \text{Cov}_M(\Delta \psi_i, \Delta \bar{\alpha}_i)}{\text{Var}_M(\Delta \psi_i + \Delta \bar{\alpha}_i)}, \quad (5)$$

a *transition* object:  $\text{Var}_M(\Delta \psi_i)$  depends on how far movers travel in the location-effect distribution, and can be small even when location effects are highly dispersed in the cross-section, if movers mostly move between similar locations; the covariance and the composition variance depend on whether movements in location effects and in composition reinforce or offset each other along realized routes.

Projecting the composition change on the location-effect change,

$$\Delta \bar{\alpha}_i = \beta_\Delta \Delta \psi_i + R_i, \quad \beta_\Delta := \frac{\text{Cov}_M(\Delta \bar{\alpha}_i, \Delta \psi_i)}{\text{Var}_M(\Delta \psi_i)}, \quad \lambda_\Delta := \frac{\text{Var}_M(R_i)}{\text{Var}_M(\Delta \psi_i)}, \quad (6)$$

and substituting into (5) gives

$$\theta_\infty = f(\beta_\Delta, \lambda_\Delta), \quad f(\beta, \lambda) := \frac{1 + \beta}{(1 + \beta)^2 + \lambda}. \quad (7)$$

The slope  $\beta_\Delta$  measures the component of composition change that moves with location-effect changes;  $\lambda_\Delta$  measures the composition change orthogonal to  $\Delta\psi_i$ , relative to the dispersion of location-effect changes. The pair  $(\beta_\Delta, \lambda_\Delta)$  is the realized *mobility anatomy*: pooled AKM variance components do not pin it down.

**A transition-network representation.** The transition moments have a compact representation on the network of realized moves. Let  $F_{jk}$  be the share of movers moving from  $j$  to  $k$ , with  $\sum_{j \neq k} F_{jk} = 1$ , and  $\bar{F}_{jk} := \frac{1}{2}(F_{jk} + F_{kj})$  the symmetrized flow. For any location-level  $f, g$ , the mass balance implied by Assumption 1 gives  $\mathbb{E}_M[\Delta f_i] = 0$ , so

$$\text{Cov}_M(\Delta f_i, \Delta g_i) = \sum_{j < k} 2\bar{F}_{jk} (f_j - f_k)(g_j - g_k) =: \mathcal{L}(f, g), \quad (8)$$

a weighted-graph Laplacian quadratic form whose edges are the moves movers make and whose weights count how often they make them;  $\theta_\infty$  is a ratio of such forms in  $(\psi, \bar{\alpha})$ .

**Proposition 1** (Undirected flow).  *$\theta_\infty$  depends on the mobility process only through the symmetrized flow  $\bar{F}$ . Two processes with the same  $\bar{F}$  but different directed flows—hence different origin-dependent destination rules and different history dependence—induce the identical estimand.*

*Proof.* In (8) the summand is symmetric in  $(j, k)$ , so only  $\bar{F}$  enters; the antisymmetric part of  $F$  cancels.  $\square$

*Remark 1* (The reduced form is without loss of generality). Proposition 1 reduces mobility to the symmetrized flow  $\bar{F}$ ; the reduction is also exhaustive. Lemma S1 in Supplemental Appendix S5 shows that for *every* admissible cross-section  $(\psi_j, \bar{\alpha}_j)$  and every symmetric flow consistent with location sizes and mover shares, there is an assignment process in the AKM class that induces it while satisfying exogenous mobility and Assumption 1. The pair (cross-section, undirected flow) is therefore a complete and unrestricted description of the environments the theory covers.

#### IV When the Mover Design Recovers the Cross-Section

Cross-sectional moments are taken over the locations movers visit: weight each location  $j$  by its mover incidence  $\mu_j$ , the share of mover endpoints located at  $j$ . This mover-incidence cross-section is the one the design can speak to.<sup>9</sup> For location-level  $f, g$ , write  $\text{Cov}_M(f, g)$  for the  $\mu$ -weighted

<sup>9</sup>This does not need to coincide with the person- or match-weighted cross-section an applied economist might be interested in. The wedge between  $\mu$ -weighted and individual-weighted benchmarks is separate from the isotropy question studied here.



covariance, and define the cross-sectional projection of composition on the location effect,

$$\bar{\alpha}_j = a_M + \beta_M \psi_j + q_j, \quad \beta_M = \frac{\text{Cov}_M(\bar{\alpha}_j, \psi_j)}{\text{Var}_M(\psi_j)}, \quad \lambda_M := \frac{\text{Var}_M(q_j)}{\text{Var}_M(\psi_j)}, \quad (9)$$

the cross-sectional analogue of the transition pair (6):  $\beta_M$  is the sorting slope and  $\lambda_M$  is the normalized dispersion of residual composition.<sup>10</sup> With  $\tilde{Y}_j := \psi_j + \bar{\alpha}_j$ , define three cross-sectional benchmarks,

$$\theta_{\text{var}} := \frac{\text{Var}_M(\psi_j)}{\text{Var}_M(\tilde{Y}_j)}, \quad \theta_{\text{sd}} := \sqrt{\theta_{\text{var}}}, \quad \theta_{\text{iso}} := \frac{\text{Cov}_M(\psi_j, \tilde{Y}_j)}{\text{Var}_M(\tilde{Y}_j)} = f(\beta_M, \lambda_M). \quad (10)$$

The first two are the variance and standard-deviation shares the applied researcher has in mind;  $\theta_{\text{iso}}$  is the coefficient from regressing the location effect on the location mean across locations.

**Theorem 1** (Recovery of the cross-section). *Maintain the AKM model, exogenous mobility, the large-location limit, and Assumption 1, with  $\text{Var}_M(\Delta\psi_i) > 0$  and  $\text{Var}_M(\psi_j) > 0$ . The following are equivalent.*

(i) **(Recovery.)** *The mover design reproduces the cross-sectional sorting projection:  $\beta_\Delta = \beta_M$  and  $\lambda_\Delta = \lambda_M$ .*

(ii) **(Directional isotropy.)** *There is a scalar  $\kappa > 0$  with*

$$\text{Var}_M(\Delta\psi_i) = \kappa \text{Var}_M(\psi_j), \quad \text{Cov}_M(\Delta\psi_i, \Delta\bar{\alpha}_i) = \kappa \text{Cov}_M(\psi_j, \bar{\alpha}_j), \quad \text{Var}_M(\Delta\bar{\alpha}_i) = \kappa \text{Var}_M(\bar{\alpha}_j).$$

*Either condition implies*

$$\theta_\infty = \theta_{\text{iso}} = (1 + \beta_M) \theta_{\text{var}}. \quad (11)$$

The proof (Appendix A) is ratio algebra. Isotropy says movers reproduce the cross-section's sorting geometry: they traverse the location-effect dimension at the same slope and cross the orthogonal composition dimension at the same intensity as the cross-section—in the network form, the Laplacian  $\mathcal{L}$  is proportional to the cross-sectional covariance operator on  $\text{span}\{\psi, \bar{\alpha}\}$ . The link is two-sided at the level of the pair  $(\beta, \lambda)$  but cannot be two-sided at the level of the scalar:  $f$  maps two coordinates to one, so distinct mobility anatomies can deliver the same  $\theta_\infty$ , and scalar equality  $\theta_\infty = \theta_{\text{iso}}$  can hold accidentally without recovery.

#### IV.1 The Mover Coefficient as a Midpoint

Write  $V_\psi := \text{Var}_M(\psi_j)$ ,  $V_\alpha := \text{Var}_M(\bar{\alpha}_j)$ ,  $C := \text{Cov}_M(\psi_j, \bar{\alpha}_j)$ ,  $V_Y := V_\psi + 2C + V_\alpha$ . Beyond the variance share  $\theta_{\text{var}} = V_\psi/V_Y$ , a distinct object asks how much cross-sectional dispersion in location means

<sup>10</sup>  $q_j$  is a linear-projection residual, not the residual from conditioning nonparametrically on  $\psi_j$ . It differs from  $v_j$  of (3) when mover shares  $w_j$  vary across locations; the two coincide when  $w_j$  is constant.



would disappear if location effects were equalized while composition were held fixed,

$$S_{\psi}^{\text{eq}} := \frac{\text{Var}_M(\bar{Y}_j) - \text{Var}_M(\bar{Y}_j - \psi_j)}{\text{Var}_M(\bar{Y}_j)} = \frac{V_{\psi} + 2C}{V_Y}. \quad (12)$$

**Corollary 1** (Midpoint interpretation and ordering). *Under directional isotropy and  $V_Y > 0$ ,*

$$\theta_{\infty} = \theta_{\text{iso}} = \frac{1}{2}(\theta_{\text{var}} + S_{\psi}^{\text{eq}}), \quad S_{\psi}^{\text{eq}} - \theta_{\infty} = \theta_{\infty} - \theta_{\text{var}} = \frac{C}{V_Y}. \quad (13)$$

*If  $C \geq 0$  then  $\theta_{\text{var}} \leq \theta_{\infty} \leq S_{\psi}^{\text{eq}}$ , and if  $C < 0$  then  $S_{\psi}^{\text{eq}} \leq \theta_{\infty} \leq \theta_{\text{var}}$ ; in either case  $S_{\psi}^{\text{eq}} \leq 1$ . Moreover  $\theta_{\infty} \leq \theta_{\text{sd}}$  for either sign of  $C$ , and if  $C \geq 0$  then  $\theta_{\infty}^2 \leq \theta_{\text{var}} \leq \theta_{\infty}$ .*

The mover coefficient recovers neither the variance share of location effects nor the full accounting effect of equalizing them; under isotropic mobility it lies exactly halfway between.

#### IV.2 Two Benchmark Readings

**Corollary 2** (The two familiar readings). *Maintain  $\text{Var}_M(\psi_j) > 0$  and  $\text{Var}_M(\bar{Y}_j) > 0$ .*

- (i) **Orthogonal composition.** *Under directional isotropy, if  $\beta_M = 0$  then  $\theta_{\infty} = \theta_{\text{iso}} = \theta_{\text{var}} = S_{\psi}^{\text{eq}}$ .*
- (ii) **Affine composition.** *If  $\lambda_M = 0$  and  $\beta_M \geq 0$ —so that  $\bar{\alpha}_j = a_M + \beta_M \psi_j$  across the locations movers visit—then every flow with  $\text{Var}_M(\Delta\psi_i) > 0$  satisfies*

$$\theta_{\infty} = \theta_{\text{sd}} = \frac{1}{1 + \beta_M}, \quad \theta_{\text{var}} = \theta_{\infty}^2, \quad S_{\psi}^{\text{eq}} = 2\theta_{\infty} - \theta_{\infty}^2.$$

Each corner has a concrete economic case. Composition is orthogonal ( $\beta_M = 0$ ) when individuals choose locations for reasons unrelated to the outcome under study and an isotropic mover design then recovers the variance share. Composition is affine ( $\lambda_M = 0$ ) when matching runs on the outcome dimension itself and the coefficient is the SD share.

The two readings differ in their robustness. The orthogonal reading relies on isotropy. The affine reading is *mobility-robust*: once composition is perfectly collinear with the location effect there is no second dimension for mover routes to sample differently, so  $\theta_{\infty} = \theta_{\text{sd}}$  for every flow. The converse fails as soon as a residual dimension exists. Once  $\lambda_M > 0$ , the mobility geometry matters and the coefficient reverts to the transition projection  $f(\beta_{\Delta}, \lambda_{\Delta})$ .

#### IV.3 Anisotropic Mobility

Outside isotropy the coefficient remains a well-defined transition object but loses the cross-sectional interpretation. Two examples illustrate the possibilities and organize the simulations.

**Example 1** (Composition-preserving mobility). If movers cross the location-effect dimension without crossing composition classes—a job ladder in which workers move to firms that hire similar coworkers, or relocation within a fixed tier of neighborhoods—then  $\Delta \bar{\alpha}_i \approx 0$  on every move,  $\beta_\Delta, \lambda_\Delta \rightarrow 0$ , and  $\theta_\infty \rightarrow 1$  regardless of the cross-sectional slope  $\beta_M > 0$ . A mover design can report full passthrough even when composition is highly dispersed across locations.

**Example 2** (Balanced mobility with anisotropic exposure). Consider four equally weighted locations indexed by  $(p, z) \in \{-1, 1\}^2$ , with  $\psi_{p,z} = p$  and  $\bar{\alpha}_{p,z} = cz$ ,  $c > 0$ , so composition is orthogonal to the location effect and  $\theta_{\text{iso}} = \theta_{\text{var}} = 1/(1 + c^2)$ . Let a balanced flow place probability  $\rho$  on moves that switch  $p$  holding  $z$  fixed, and  $1 - \rho$  on moves that switch  $z$  holding  $p$  fixed. Then  $\theta_\infty = \rho/[\rho + c^2(1 - \rho)]$ : at  $\rho = \frac{1}{2}$  movers expose the two dimensions in the cross-sectional proportion and  $\theta_\infty = \theta_{\text{var}}$ ; for  $\rho < \frac{1}{2}$  they over-sample residual composition and  $\theta_\infty < \theta_{\text{var}}$ ; as  $\rho \rightarrow 1$  they avoid it and  $\theta_\infty \rightarrow 1$ . Balanced inflows and outflows alone do not deliver a cross-sectional reading; what matters is the geometry of the realized routes.

## V Testing the Cross-Sectional Interpretation

Theorem 1 turns the cross-sectional interpretation into two testable restrictions. With  $Q_M := \text{Var}_M(q_j)$ ,  $Q_\Delta := \text{Var}_M(R_i)$ ,  $V_{\psi,M} := \text{Var}_M(\psi_j)$ , and  $V_{\psi,\Delta} := \text{Var}_M(\Delta\psi_i)$ , directional isotropy is equivalent to

$$g_\beta := \beta_\Delta - \beta_M = 0, \quad g_Q := Q_\Delta V_{\psi,M} - Q_M V_{\psi,\Delta} = 0. \quad (14)$$

The first asks whether movers traverse the cross-sectional sorting slope; the second whether they cross the residual-composition dimension with the same normalized intensity. The null is that the transition and cross-sectional projection pairs coincide.

Location effects and means are estimated with finite-cell noise, and squaring noisy estimates inflates second moments—the same plug-in inflation that drives limited-mobility bias in sparse networks (Andrews et al., 2008; Kline et al., 2020); in the dense regime here it is a finite-sample nuisance, which cross-fitting removes exactly. I therefore split the workers into two random halves, estimate the first stage separately within each half  $s \in \{1, 2\}$ — $\hat{\psi}_j^s$  from the half’s movers,  $\hat{Y}_j^s$  from its cell means—and form every quadratic moment as a symmetrized *cross-split* product, one factor from each half, so that the two uncorrelated estimation errors multiply to mean zero. Six such cross-fitted primitive moments deliver  $\hat{g} = (\hat{g}_\beta, \hat{g}_Q)'$  through smooth maps. Because every first-stage estimator is linear in outcomes, each worker’s influence on  $\hat{g}$ —including the dependence among movers sharing origins and destinations—is available in closed form, and a worker-level multiplier bootstrap requires no re-estimation (Supplemental Appendix S4).

**Proposition 2** (Inference). Fix  $J$  and let  $N \rightarrow \infty$  with  $n_j/N \rightarrow s_j > 0$  for every  $j$  (so  $n_j \rightarrow \infty$  at rate  $N$ ), let workers be independent with uniformly bounded fourth moments of outcomes,  $\sup_{i,t} \mathbb{E}[Y_{it}^4] < \infty$ , and both split mover graphs connected with probability approaching one, let  $\text{Var}_M(\psi_j)$  and  $\text{Var}_M(\Delta\psi_i)$  be bounded away from zero, and let the asymptotic covariance  $\Omega$  of  $\sqrt{N}(\hat{g} - g)$ , characterized in Supplemental Appendix S4,

be nonsingular. Then  $\sqrt{N}(\widehat{g} - g) \xrightarrow{d} \mathcal{N}(0, \Omega)$ , the worker-level multiplier bootstrap delivers  $\widehat{\Omega}_g$  consistent for the finite-sample covariance  $N^{-1}\Omega$  of  $\widehat{g}$ , and under  $H_0: g = 0$  the Wald statistic  $\mathcal{W} = \widehat{g}'\widehat{\Omega}_g^{-1}\widehat{g} \xrightarrow{d} \chi_2^2$ .

$\widehat{g}_\beta$  shows whether mover routes load on a different sorting slope than the cross-section,  $\widehat{g}_Q$  whether they over- or under-sample residual composition. Under the null, the midpoint reading (13) applies; under rejection, the coefficient remains a valid transition estimand but cannot be mapped to a cross-sectional share, and  $(\widehat{g}_\beta, \widehat{g}_Q)$  reveal which failure drives the wedge.<sup>11</sup>

## VI Simulations

I evaluate the theory on a simulated two-period AKM economy,  $Y_{it} = \alpha_i + \psi_{j(i,t)} + \varepsilon_{it}$ . The cross-section is fixed by design:  $J = 256$  locations with unit-variance place effects and composition  $\bar{\alpha}_j = \beta_M \psi_j + q_j$ , sorting slope  $\beta_M = 0.6$ , residual intensity  $\lambda_M = 0.7$ ; the orthogonal and affine designs set  $\beta_M = 0$  and  $\lambda_M = 0$ , respectively. Each location has a stayer core and one-time movers ( $\approx 332,000$  workers, mover share 7.5 percent, within-location dispersion of  $\alpha_i$  equal to one,  $\sigma_\varepsilon = 0.2$ ). Stayers carry  $\bar{\alpha}_j$ ; mover types are centred on the symmetric edge potentials of the representation lemma (Supplemental Appendix S5), so leavers, arrivers, and stayers at every location share the same mean composition in both periods and Assumption 1 holds exactly under every flow. The cross-section—and with it the pooled AKM decomposition—is thus a constant of the economy; only the routes differ.

Mobility enters only through the symmetrized flow. Random reshuffling draws destinations independently of origins and reproduces the cross-sectional geometry exactly,  $\beta_\Delta = \beta_M$  and  $\lambda_\Delta = \lambda_M$ : directional isotropy. The anisotropic designs mix the two move types of Example 2—with probability  $\rho$  a job-ladder step, crossing the place-effect dimension at fixed coworker type; with probability  $1 - \rho$  a career switch, crossing coworker type at fixed place effect—which holds  $\beta_\Delta = \beta_M$  and traces  $\lambda_\Delta = \frac{1-\rho}{\rho}\lambda_M$ : isotropic at  $\rho = \frac{1}{2}$ , oversampling the residual dimension for  $\rho < \frac{1}{2}$ , undersampling it for  $\rho > \frac{1}{2}$ . Each of 500 replications draws the individual-level panel and runs the Section V procedure end to end: split-sample AKM on the largest connected set of each half, cross-fitted moments, the leave-one-out mover regression, and the worker-level bootstrap test.

Table 1 reports three findings. First, under isotropic mobility the mover coefficient equals the cross-sectional midpoint of Corollary 1 (0.49), distinct from both the variance share (0.31) and the SD share (0.55); the orthogonal and affine rows reproduce the two corner readings of Corollary 2. Second—the headline—the four Panel B economies share an *identical* cross-section and an essentially identical pooled AKM decomposition, so an AKM variance study would call them one economy; yet the mover coefficient ranges from 0.25 to 0.61 purely because movers traverse the same map differently. As the flow tilts from career switches ( $\rho = 0.15$ ,  $\widehat{\lambda}_\Delta \approx 4$ ) toward near-pure job-ladder steps ( $\rho = 0.92$ ,  $\widehat{\lambda}_\Delta \rightarrow 0$ ),  $\widehat{\theta}$  sweeps from far below  $\theta_{\text{iso}}$  to well above it, while every

<sup>11</sup>When composition is exactly affine ( $\lambda_M = \lambda_\Delta = 0$ ), the parameter sits on the boundary of the parameter space and the  $\chi_2^2$  reference is conservative; the operative hypothesis there is the one-sided  $\lambda_\Delta = \lambda_M = 0$ , and by Corollary 2(ii) the SD-share reading is mobility-robust anyway. See Supplemental Appendix S4.

cross-sectional benchmark— $\theta_{\text{iso}}$ ,  $\theta_{\text{var}}$ ,  $\lambda_M$ —stays put. Third, the isotropy test holds nominal size in the interior isotropic economies, is conservative at the affine boundary where the SD-share reading is mobility-robust in any case, and rejects every anisotropic flow with probability one; Panel C confirms the robustness—an affine cross-section under a residual-heavy flow still delivers  $\hat{\theta} = \theta_{\text{sd}}$  and no rejection.

## VII Empirical Illustration: Dutch Administrative Data

I use the Statistics Netherlands (CBS) linked employer–employee panel covering 2009–2016. The outcome is the log gross hourly wage<sup>12</sup> of full-time male workers, residualized on age and calendar-year indicators in a preliminary step; the Supplemental Appendix repeats the analysis on the unadjusted wage with the same conclusions. Following the two-period convention of Section II, I retain stayers and one-time movers, and I drop firms below the 25th percentile of total worker-period observations. Moves are inferred from the firm identifier: a one-time mover changes employer exactly once, the move year being the first year observed at the destination. The worker’s origin and destination spells are the two periods of Section II— $\Delta Y_i$  is the worker’s average outcome across destination years minus the average across origin years—while  $\delta_i^{(-i)}$  follows the Finkelstein et al. (2016) simple-year-average convention: leave-one-out firm means averaged over the sample window, destination minus origin. Averaging over the window is deliberate: firm composition drifts across 2009–2016, so no single year defines *the* cross-section; the benchmark tested is the time-averaged one—the object the leave-one-out regressor evaluates. The estimation sample comprises 203,508 one-time movers and 992,254 stayers at 8,339 firms, with a mean firm-year size of 130 workers; full counts are in the notes to Table 2.

Workers are split into two random halves; within each, AKM recovers  $\hat{\psi}_j^s$  on the largest connected component of mover transitions, and firm-year means use all workers in the half. Diagnostics are evaluated on the intersection of the two connected sets, which retains 97.8 percent of movers; inference uses the worker-level multiplier bootstrap of Supplemental Appendix S4 with 1,000 draws.

Table 2 reports the results. The mover-design coefficient is  $\hat{\theta} = 0.214$ , and the transition projection nearly reproduces it ( $f(\hat{\beta}_\Delta, \hat{\lambda}_\Delta) = 0.203$ ).<sup>13</sup> The  $\mu$ -weighted benchmarks are tightly estimated:  $\theta_{\text{var}} = 0.259$ ,  $\theta_{\text{sd}} = 0.508$ ,  $S_\psi^{\text{eq}} = 0.283$ , and  $\theta_{\text{iso}} = 0.271$ —the midpoint of Corollary 1 exactly.

Panel B reports the sorting geometry, and it fails isotropy on both coordinates. The cross-sectional sorting slope is statistically indistinguishable from zero ( $\hat{\beta}_M = 0.047$ ), while the transition slope is

<sup>12</sup>The hourly wage is annual gross contract earnings divided by contracted annual hours, computed as the job’s full-time factor (FTE) times a 52-week, 40-hour full-time year; the sample is restricted to FTE 0.9.

<sup>13</sup>Finite-cell noise in the leave-one-out regressor attenuates  $\hat{\theta}$ , so under the static model the regression sits *below* the cross-fitted projection. The small positive gap here is instead the signature of the post-event drift of Figure 1: the post-window average embeds returns accruing beyond the move year, lifting the regression slightly above the static-model projection.

Table 1: Simulated Mover Designs: the Same AKM Decomposition Can Hide Different Estimands

| Design                                                                                              | Pooled AKM decomposition |            |      | Cross-sectional readings |                       |                       |                      |                        |                                                                     | Residual share    |                        |                    |
|-----------------------------------------------------------------------------------------------------|--------------------------|------------|------|--------------------------|-----------------------|-----------------------|----------------------|------------------------|---------------------------------------------------------------------|-------------------|------------------------|--------------------|
|                                                                                                     | Var $\alpha$             | Var $\psi$ | Corr | $\hat{\theta}$           | $\theta_{\text{iso}}$ | $\theta_{\text{var}}$ | $\theta_{\text{sd}}$ | $S_{\psi}^{\text{eq}}$ | midpt.<br>$\frac{1}{2}(\theta_{\text{var}} + S_{\psi}^{\text{eq}})$ | $\hat{\lambda}_M$ | $\hat{\lambda}_\Delta$ | Rej.<br>$H_0$      |
| <b>Panel A: Random reshuffling (directionally isotropic); vary the cross-section</b>                |                          |            |      |                          |                       |                       |                      |                        |                                                                     |                   |                        |                    |
| General ( $\beta > 0, \lambda_M > 0$ )                                                              | 2.14                     | 1.00       | 0.41 | 0.491                    | 0.491                 | 0.307                 | 0.554                | 0.675                  | 0.491                                                               | 0.701             | 0.700                  | 0.054              |
| Orthogonal ( $\beta = 0$ )                                                                          | 1.75                     | 1.00       | 0.00 | 0.588                    | 0.588                 | 0.588                 | 0.767                | 0.588                  | 0.588                                                               | 0.700             | 0.700                  | 0.040              |
| Affine ( $q_j \equiv 0$ )                                                                           | 1.39                     | 1.00       | 0.51 | 0.625                    | 0.625                 | 0.391                 | 0.625                | 0.859                  | 0.625                                                               | 0.000             | 0.000                  | 0.020 <sup>a</sup> |
| <b>Panel B: Same cross-section and same AKM decomposition; vary only mobility <math>\rho</math></b> |                          |            |      |                          |                       |                       |                      |                        |                                                                     |                   |                        |                    |
| Balanced ( $\rho = 0.50$ )                                                                          | 2.09                     | 1.00       | 0.42 | 0.491                    | 0.491                 | 0.307                 | 0.554                | 0.675                  | 0.491                                                               | 0.700             | 0.699                  | 0.048              |
| Job ladder, $\psi$ -heavy ( $\rho = 0.85$ )                                                         | 2.09                     | 1.00       | 0.42 | <b>0.596</b>             | 0.491                 | 0.307                 | 0.554                | 0.675                  | 0.491                                                               | 0.699             | 0.123                  | 1.000              |
| Career switching, residual-heavy ( $\rho = 0.15$ )                                                  | 2.11                     | 1.00       | 0.41 | <b>0.245</b>             | 0.491                 | 0.307                 | 0.554                | 0.675                  | 0.491                                                               | 0.701             | 3.969                  | 1.000              |
| Job ladder, near-pure ( $\rho = 0.92$ )                                                             | 2.09                     | 1.00       | 0.42 | <b>0.610</b>             | 0.491                 | 0.307                 | 0.554                | 0.675                  | 0.491                                                               | 0.700             | 0.061                  | 1.000              |
| <b>Panel C: Affine composition is mobility-robust</b>                                               |                          |            |      |                          |                       |                       |                      |                        |                                                                     |                   |                        |                    |
| Affine + residual-heavy flow                                                                        | 1.36                     | 1.00       | 0.51 | 0.624                    | 0.625                 | 0.391                 | 0.625                | 0.859                  | 0.625                                                               | 0.000             | 0.000                  | 0.002 <sup>a</sup> |

*Notes.* Monte Carlo means over 500 replications of the individual-level economy:  $J = 256$  locations,  $\approx 1,200$  stayers per location and 25,000 movers ( $\approx 332,000$  workers, mover share 7.5%),  $Y_{it} = \alpha_i + \psi_{i(i,t)} + \varepsilon_{it}$  with  $\sigma_\alpha = 1$  (the within-location dispersion of  $\alpha_i$ ; the total Var  $\alpha$  adds the between-location composition),  $\sigma_\varepsilon = 0.2$ ,  $\beta_M = 0.6$ ,  $\lambda_M = 0.7$ . Mover effects are pinned on symmetric edge potentials so that each location's composition is identical in both periods (Section VI). Each replication estimates  $\hat{\psi}_i$  by AKM on the largest connected set within each of two worker splits, forms leave-one-out coworker means, cross-fits the six  $\mu$ -weighted primitive moments, and applies the 1,000-draw worker-level multiplier bootstrap of Supplemental Appendix S4, propagating first-stage  $\hat{\psi}$  and cell-mean noise in closed form. The first three columns are the pooled population AKM moments;  $\hat{\theta}$  is the leave-one-out mover-regression coefficient (4); the midpoint column verifies  $\theta_{\text{iso}} = \frac{1}{2}(\theta_{\text{var}} + S_{\psi}^{\text{eq}})$ . "Rej.  $H_0$ " is the 5% rejection frequency of the joint isotropy test: the test's size in the isotropic rows and its *power* in the anisotropic rows. In Panel B the cross-section—and hence every cross-sectional reading and  $\lambda_M$ —is held *exactly* fixed across rows; only  $\rho$  changes, and the pooled Var  $\alpha$  and Corr coincide up to about a percent.

<sup>a</sup> At the affine boundary  $\lambda_M = \lambda_\Delta = 0$  the parameter sits on the edge of the parameter space and the  $\chi^2_2$  reference is conservative (Remark S1); the test under-rejects rather than spuriously rejecting the one-dimensional configuration, where the SD-share reading is mobility-robust in any case (Corollary 2). The interior rows show correct size.

negative ( $\hat{\beta}_\Delta = -0.075$ ): movers who gain firm premium move toward slightly lower-composition coworkers. Residual composition dominates the cross-section itself ( $\hat{\lambda}_M = 2.77$ ), so the affine benchmark of Corollary 2(ii) is rejected outright and the mobility-robust SD-share reading is unavailable; movers oversample the residual dimension by a third ( $\hat{\lambda}_\Delta = 3.71$ ). Both restrictions reject individually, and the joint Wald statistic is 135.6 (bootstrap  $p < 0.001$ ).

This conclusion invokes no stationarity: by Supplemental Appendix S3, with the cross-section read as the time-averaged object above, the diagnostic compares two sets of sample moments from one data set. Its rejection is a property of the data, not of a maintained steady state—the realized routes traverse a different sorting geometry than the cross-section they connect.

The interpretive consequences are exactly those of Section IV. With isotropy rejected,  $\hat{\theta}$  is the transition projection and inherits no cross-sectional reading. The mobility pattern matches the career-switching anatomy of Example 2 and the residual-heavy row of Table 1: heavy traverse of residual composition pushes the coefficient below every cross-sectional benchmark.

Figure 1 reports the multi-period FGW scaled event study. Pre-event coefficients are economically negligible—at most 0.015 in absolute value—though statistically distinguishable from zero at this sample size. Post-event coefficients jump to 0.123 in the move year, mechanically attenuated because the move year straddles both firms, and drift to 0.206 by  $r = 4$ —a mild departure from the static model, consistent with on-the-job returns at the destination. The drift affects  $\hat{\theta}$  directly, because the post-event window averages over it. The test does not use movers’ wage changes; it is computed from the estimated firm effects. The drift can therefore affect the test only by biasing those estimates.<sup>14</sup>

## VIII Conclusion

Mover-design event study coefficients identify a transition-weighted projection on changes in location means, not a generic share of the cross-sectional variance of location means due to places. The projection depends on mobility only through the undirected flow between locations, and it admits a cross-sectional interpretation exactly when mobility reproduces the cross-section’s sorting geometry—a testable condition. When the condition holds, the recovered object is the midpoint between the variance share of place effects and the equalization share; the standard-deviation-share reading is the knife-edge case of affine sorting, and it is the only such reading robust to arbitrary mobility. When the condition fails—as it does in the Dutch wage-data illustration, where movers traverse far more residual composition than the cross-section contains—the coefficient cannot be mapped to a cross-sectional share, and pooled AKM moments cannot detect the failure. The practical prescription is correspondingly simple: report the transition pair  $(\hat{\beta}_\Delta, \hat{\lambda}_\Delta)$  alongside the mover coefficient, test the isotropy restrictions before attaching a cross-sectional variation interpretation to it.

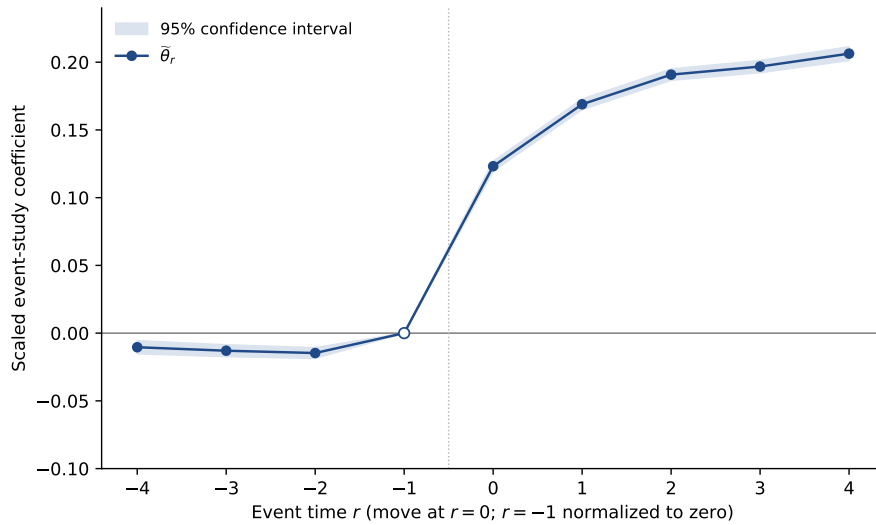
<sup>14</sup>For the drift to explain the rejection, post-move wage growth would have to bias the estimated firm effects substantially, and differently for different kinds of moves. The rejection is best read as a result within the static AKM specification; the illustration demonstrates the diagnostic, it does not offer a definitive decomposition of Dutch wage mobility.

Table 2: Mover-Design Diagnostics, Dutch Administrative Data 2009–2016

|                                                                         | Estimate | SE    | 95% CI           |
|-------------------------------------------------------------------------|----------|-------|------------------|
| <i>Panel A. Mover-design coefficient and cross-sectional benchmarks</i> |          |       |                  |
| Mover regression $\hat{\theta}$                                         | 0.214    | 0.002 | [0.210, 0.218]   |
| Transition projection $f(\hat{\beta}_\Delta, \hat{\lambda}_\Delta)$     | 0.203    | 0.004 | [0.195, 0.211]   |
| $\theta_{\text{iso}}$ (midpoint)                                        | 0.271    | 0.007 | [0.257, 0.285]   |
| $\theta_{\text{var}}$                                                   | 0.259    | 0.015 | [0.230, 0.288]   |
| $\theta_{\text{sd}}$                                                    | 0.508    | 0.014 | [0.480, 0.537]   |
| $S_\psi^{\text{eq}}$                                                    | 0.283    | 0.014 | [0.253, 0.310]   |
| <i>Panel B. Sorting geometry: cross-section versus transitions</i>      |          |       |                  |
| Cross-sectional slope $\hat{\beta}_M$                                   | 0.047    | 0.052 | [−0.045, 0.156]  |
| Transition slope $\hat{\beta}_\Delta$                                   | −0.075   | 0.029 | [−0.127, −0.010] |
| Cross-sectional residual intensity $\hat{\lambda}_M$                    | 2.772    | 0.134 | [2.522, 3.053]   |
| Transition residual intensity $\hat{\lambda}_\Delta$                    | 3.712    | 0.136 | [3.458, 3.994]   |
| <i>Panel C. Isotropy test</i>                                           |          |       |                  |
| Slope restriction $\hat{g}_\beta$                                       | −0.122   | 0.045 | [−0.210, −0.033] |
| Residual restriction $\hat{g}_Q (\times 10^4)$                          | 1.089    | 0.181 | [0.748, 1.451]   |
| Joint Wald $\mathcal{W} (\chi^2_2 p = 3.6 \times 10^{-30})$             | 135.566  | —     | —                |

*Notes.* CBS linked employer–employee data, 2009–2016, full-time male workers; outcome is log gross hourly wage residualized on age and calendar-year indicators. Stayers and one-time movers; firms below the 25th percentile of worker-period observations dropped.  $N = 7,966,160$  worker-years, 1,195,762 workers, 992,254 stayers and 203,508 one-time movers (a worker-level mover share of 17.0 percent), 8,339 firms; diagnostics on the intersection of the two split connected sets, which retains 6,391 firms (76.7 percent) and 198,955 movers (97.8 percent), of whom 198,846 have leave-one-out cell means. SEs and percentile CIs from the worker-level multiplier bootstrap of Supplemental Appendix S4 with 1,000 draws; the SE for  $\hat{\theta}$  is the conventional regression SE, and its CI the corresponding analytic normal interval.  $\theta_{\text{iso}}$  equals  $\frac{1}{2}(\theta_{\text{var}} + S_\psi^{\text{eq}})$  exactly, and the two formulas for the transition projection agree by construction; both identities hold in the estimates as reported. The bootstrap  $p$ -value of the joint Wald is below 0.001.

Figure 1: Event Study of Outcomes around the Mover Transition



*Notes.* Per-horizon coefficients  $\hat{\theta}_r$  from the long-differenced multi-period mover-design regression of Supplemental Appendix S1, with base period  $r = -1$  normalized to zero, estimated on the mover sample (1,290,176 mover-year observations; sample as in Table 2). The shaded band is the pointwise 95 percent analytical confidence interval.



## A Proofs

**Proof of Theorem 1.** (ii) $\Rightarrow$ (i): dividing the three displayed equalities,  $\beta_\Delta = \text{Cov}_M(\Delta\psi_i, \Delta\bar{\alpha}_i)/\text{Var}_M(\Delta\psi_i) = \text{Cov}_M(\psi_j, \bar{\alpha}_j)/\text{Var}_M(\psi_j) = \beta_M$ , and  $\lambda_\Delta = \text{Var}_M(\Delta\bar{\alpha}_i)/\text{Var}_M(\Delta\psi_i) - \beta_\Delta^2 = \text{Var}_M(\bar{\alpha}_j)/\text{Var}_M(\psi_j) - \beta_M^2 = \lambda_M$ . (i) $\Rightarrow$ (ii): set  $\kappa := \text{Var}_M(\Delta\psi_i)/\text{Var}_M(\psi_j) > 0$ ; then  $\beta_\Delta = \beta_M$  delivers the cross-covariance equality and  $\lambda_\Delta = \lambda_M$  the  $\bar{\alpha}$  equality. For (11), divide the numerator and denominator of (5) by  $\text{Var}_M(\Delta\psi_i)$  to obtain  $\theta_\infty = f(\beta_\Delta, \lambda_\Delta)$ ; under (i) this is  $f(\beta_M, \lambda_M) = \theta_{\text{iso}} = (1 + \beta_M)\theta_{\text{var}}$ , using  $\text{Cov}_M(\psi_j, \bar{Y}_j) = \text{Var}_M(\psi_j)(1 + \beta_M)$  and  $\text{Var}_M(\bar{Y}_j) = \text{Var}_M(\psi_j)[(1 + \beta_M)^2 + \lambda_M]$ .  $\square$

**Proof of Corollary 1.** Under isotropy, Theorem 1 gives  $\theta_\infty = \theta_{\text{iso}} = (V_\psi + C)/V_Y$ . With  $\theta_{\text{var}} = V_\psi/V_Y$  and  $S_\psi^{\text{eq}} = (V_\psi + 2C)/V_Y$ , the average of the two is  $(V_\psi + C)/V_Y = \theta_\infty$ ; subtracting gives the distance identity. If  $C \geq 0$  the ordering follows from  $1 - S_\psi^{\text{eq}} = V_\alpha/V_Y \geq 0$ ; the case  $C < 0$  is analogous. For the secondary bounds: when  $1 + \beta_M > 0$ ,  $\theta_\infty/\theta_{\text{sd}} = (1 + \beta_M)\sqrt{\theta_{\text{var}}} \leq 1$  is equivalent to  $\lambda_M \geq 0$ , and the bound is immediate when  $1 + \beta_M \leq 0$ ; and if  $C \geq 0$ ,  $\theta_\infty^2 \leq \theta_{\text{var}} \leq \theta_\infty$  follows from the midpoint identity by substitution.  $\square$

**Proof of Corollary 2.** (i)  $\beta_M = 0$  gives  $C = 0$ ; isotropy then yields  $\theta_\infty = \theta_{\text{iso}} = V_\psi/V_Y = \theta_{\text{var}} = S_\psi^{\text{eq}}$ . (ii)  $\lambda_M = 0$  forces  $\text{Var}_M(q_j) = 0$ , so  $\bar{\alpha}_j = a_M + \beta_M\psi_j$  at every location movers visit and every move obeys  $\Delta\bar{\alpha}_i = \beta_M\Delta\psi_i$ ,  $\Delta\bar{Y}_i = (1 + \beta_M)\Delta\psi_i$ ; hence for any flow with  $\text{Var}_M(\Delta\psi_i) > 0$ ,  $\theta_\infty = 1/(1 + \beta_M) = \theta_{\text{sd}}$ , and the remaining identities follow by substitution.  $\square$

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## Supplemental Appendix

### When Is the Mover-Design Event Study Coefficient a Place-Effect Share?

Vahid Moghani

This Supplemental Appendix establishes the equivalence of the multi-period [Finkelstein et al. \(2016\)](#) event study with the two-period first difference used in the paper (Section [S1](#)), characterizes the flow restrictions implied by stationarity (Section [S2](#)), extends the recovery results to the time-averaged cross-section when stationarity is dropped (Section [S3](#)), develops the asymptotic theory and multiplier bootstrap behind the isotropy test (Section [S4](#)), proves a representation lemma showing that the paper’s reduced-form objects exhaust the AKM class (Section [S5](#)), and reports descriptive statistics for the estimation sample (Section [S6](#)) and the robustness of the empirical results to the unadjusted log hourly wage (Section [S7](#)). It is referenced from, but—except for the inference theory of Section [S4](#)—not required for, the results in the paper.

#### S1 The FGW Event Study and Two-Period Equivalence

The applied mover-design event study is the multi-period [Finkelstein et al. \(2016\)](#) (FGW) specification with base period  $k = -1$ . This section makes its relation to the paper’s two-period objects exact, in two steps. First, on a balanced event-time window the FGW coefficients coincide, by projection algebra alone, with a sequence of per-event-time long-difference regressions (Proposition [S1](#)). Second, under the AKM outcome model with residual exogeneity, those per-horizon coefficients converge—pre-event to zero, post-event to a common limit, the two-period first-difference estimand (Theorem [S1](#)).

Consider movers  $i \in M$  observed in periods  $t \in \{1, \dots, T\}$ ,  $T \geq 4$ , each changing location exactly once at  $t_i^*$ , with event time  $\tau_{it} := t - t_i^*$  and  $o(i) := j(i, t_i^* - 1)$ ,  $d(i) := j(i, t_i^*)$ .

**Assumption S1** (AKM residual exogeneity).  $Y_{it} = \alpha_i + \psi_{j(i,t)} + \varepsilon_{it}$  with  $\psi_{j(i,t)} = \psi_{o(i)}$  for  $t < t_i^*$  and  $\psi_{d(i)}$  thereafter;  $\mathbb{E}[\varepsilon_{it} \mid \mathcal{F}^T] = 0$  where  $\mathcal{F}^T$  is generated by individual effects, location effects, the realized event-time assignment network, and the non-idiosyncratic components of coworker composition; residuals conditionally uncorrelated across  $i$  and across  $t$  within  $i$  ([Card et al., 2013, 2018](#)).

**Assumption S2** (Balanced event-time window). Fix  $\mathcal{K} := \{-K_-, \dots, -2\} \cup \{0, \dots, K_+\}$  and restrict move dates to  $K_- + 1 \leq t_i^* \leq T - K_+$ , so that every mover’s event window  $\mathcal{K} \cup \{-1\}$  lies inside the calendar panel. Estimation uses only the event-window observations,  $\tau_{it} \in \mathcal{K} \cup \{-1\}$ .

Following equation (6) of [Finkelstein et al. \(2016\)](#)<sup>15</sup>, the event-study specification is

$$Y_{it} = \tilde{a}_i + \theta_{r(i,t)} \delta_i^{(-i)} + \rho_{r(i,t)} + e_{it}, \quad \delta_i^{(-i)} := \bar{Y}_{d(i),t_i^*}^{(-i)} - \bar{Y}_{o(i),t_i^*-1}^{(-i)}, \quad (\text{S1})$$

estimated by OLS on the mover sample over the window of Assumption S2: individual effects  $\tilde{a}_i$ , event-time main effects  $\rho_r$ , and the leave-one-out destination–origin gap  $\delta_i^{(-i)}$  interacted with event time, normalized at the base period ( $\theta_{-1} = \rho_{-1} = 0$ ).<sup>16</sup> The coefficients of interest are the  $\theta_k$ .

The reference point is the per- $k$  long difference: for  $k \in \mathcal{K}$  define  $Y_{ik}^* := Y_{i,t_i^*+k} - Y_{i,t_i^*-1}$  and estimate  $Y_{ik}^* = c_k + \tilde{\theta}_k \delta_i^{(-i)} + e_{ik}$  by OLS, so that  $\tilde{\theta}_k$  is the centered slope  $\text{Cov}_M(\delta_i^{(-i)}, \cdot) / \text{Var}_M(\delta_i^{(-i)})$  used below.<sup>17</sup> These per- $k$  long differences are the canonical estimator throughout the paper; the next result shows the FGW coefficients coincide with them.

**Proposition S1** (The FGW coefficients are per- $k$  long differences). *In the population projection corresponding to equation (S1) on the balanced window of Assumption S2,  $\theta_k = \tilde{\theta}_k$  and  $\rho_k = c_k$  for every  $k \in \mathcal{K}$ .*

*Proof.* Partial out the individual effects. On the balanced window, within-mover demeaning acts identically across movers, and the specification is saturated in event time—main effects and interactions at every  $k$ —so the projection separates into the per- $k$  contrasts of event time  $k$  against the base period. Each contrast is the regression of  $Y_{ik}^*$  on  $(1, \delta_i^{(-i)})$ , delivering  $\rho_k = c_k$  and  $\theta_k = \tilde{\theta}_k$ .  $\square$

**Theorem S1.** *Under Assumptions S1 and S2, in the fixed- $J$ , large-location regime of the paper, as  $N \rightarrow \infty$ : (i) for every  $k \leq -2$ ,  $\tilde{\theta}_k \xrightarrow{p} 0$ ; (ii) for every  $k \geq 0$ ,  $\tilde{\theta}_k \xrightarrow{p} \theta_{\text{FD}} := \text{Cov}_M(\Delta\psi_i, \delta_i^{(-i)}) / \text{Var}_M(\delta_i^{(-i)})$ , independent of  $k$ .*

*Proof.* Under Assumption S1,  $Y_{it} = a_i + \Delta\psi_i \mathbf{1}\{\tau_{it} \geq 0\} + \varepsilon_{it}$  with  $a_i := \alpha_i + \psi_{o(i)}$ , so  $Y_{ik}^* = \Delta\psi_i \mathbf{1}\{k \geq 0\} + \Delta\varepsilon_{ik}$  with  $\Delta\varepsilon_{ik} := \varepsilon_{i,t_i^*+k} - \varepsilon_{i,t_i^*-1}$ . By the law of large numbers,  $\tilde{\theta}_k \rightarrow [\mathbf{1}\{k \geq 0\} \text{Cov}_M(\Delta\psi_i, \delta_i^{(-i)}) + \text{Cov}_M(\Delta\varepsilon_{ik}, \delta_i^{(-i)})] / \text{Var}_M(\delta_i^{(-i)})$ . Expanding  $\delta_i^{(-i)}$  into composition, location-effect, and error-mean components, the mean independence and conditional uncorrelatedness of Assumption S1 give  $\text{Cov}_M(\Delta\varepsilon_{ik}, \delta_i^{(-i)}) = 0$ . The post-event limit does not depend on  $k$  because  $\psi_j$  is time-invariant.  $\square$

<sup>15</sup>FGW’s specification additionally includes calendar-time effects and age controls; both are covered by the time-varying-controls remark below. Estimating calendar effects *jointly* within equation (S1) requires *cohort exchangeability*—that the joint distribution of  $(\Delta\psi_i, \delta_i^{(-i)})$  not vary with the move date  $t_i^*$ : after partialling the individual effects, calendar dummies are functions of  $(k, t_i^*)$ , so without exchangeability they load on cohort-varying moments of  $(\Delta\psi_i, \delta_i^{(-i)})$  and the  $\theta_k$  mix cohorts in the manner of the staggered-adoption critiques ([Sun and Abraham, 2021](#); [De Chaisemartin and d’Haultfoeuille, 2020](#); [Borusyak et al., 2024](#)). The per- $k$  long differences remain valid cohort by cohort either way, and Theorem S1 applies within each cohort.

<sup>16</sup>FGW omit move-year observations, in which a mover splits the year between origin and destination, so their post-event window effectively starts at  $k = 1$ ; the implementation in Section VII of the paper retains the move year, which is why Figure 1 shows the attenuated  $r = 0$  coefficient.

<sup>17</sup>Under the period-by-period analogue of the mass balance implied by Assumption 1 of the paper,  $\mathbb{E}_M[\delta_i^{(-i)}] = 0$  and  $c_k \xrightarrow{p} 0$ , so the intercept is asymptotically idle; carrying it makes the centered limit hold with no balance condition, consistent with the no-stationarity reading of Section S3.

**Time-varying controls.** Applied AKM specifications include time-varying controls,  $Y_{it} = \alpha_i + \psi_{j(i,t)} + X'_{it}\gamma + \varepsilon_{it}$ , with calendar-time effects the leading case. Every object in the paper is then defined on the control-adjusted outcome  $\tilde{Y}_{it} := Y_{it} - X'_{it}\gamma$ : location means, the mover regression, the transition and cross-sectional moments, and the benchmark shares. The interpretation adjusts accordingly—the recovered shares are shares of the variance of *control-adjusted* location means (age- and year-adjusted wages in the application of Section VII of the paper)—and Theorem S1 holds unchanged with  $\tilde{Y}$  in place of  $Y$ .

## S2 Stationarity and Restrictions on Mover Flows

Assumption 1 of the paper requires the average mover composition at each location to be the same whether viewed as an origin or a destination. Let  $f_{jk}$  be the mover flow from  $j$  to  $k$ ,  $\mu_{jk}$  the average individual effect along that flow, and  $\Phi_{jk} := f_{jk}\mu_{jk}$  the effect-weighted flow. Constant location sizes and mover shares give mass balance  $\sum_k f_{jk} = \sum_l f_{lj}$  at every  $j$ ; combining with the composition condition yields

$$\sum_k \Phi_{jk} = \sum_l \Phi_{lj}, \quad \forall j : \quad (\text{S2})$$

the total individual effect flowing out of a location must equal the total flowing in. Conditional on balanced masses this imposes at most  $J - 1$  independent linear restrictions on the effect-weighted flows, leaving origin–destination pairings and within-pair sorting otherwise free. In the special case where destination assignment is independent of  $\alpha_i$  conditional on origin, (S2) forces the average mover type to be constant within each connected component of the mover network; with dependence between mobility and  $\alpha_i$ , a large class of mobility patterns satisfies stationarity while generating very different mover-design estimands. Flows satisfying mass balance and (S2) are the *circulations*: the mobility that leaves the cross-section unchanged. A general flow splits into such a circulation and a residual that carries the composition drift; Assumption 1 keeps the circulation, and Section S3 shows that restoring the residual costs the results nothing.

## S3 Recovery Without Stationarity

Assumption 1 is a convenience, not a premise of the design. It does two things in the main text: it identifies each location’s period-1 and period-2 cross-sections as a single object  $\bar{\alpha}_j$ , and—through the mass balance  $\mathbb{E}_M[\Delta f_i] = 0$ —it reduces the directed flow to the symmetrized flow  $\bar{F}$  of Proposition 1. This section shows that neither role is needed for the estimand of Section III, the recovery characterization of Theorem 1, or the test of Section V. Stationarity is required *only* for the undirected-flow representation, in which the direction and order of moves wash out; the diagnostic itself never uses it. Equivalently: a drifting flow mixes the cross-section-preserving circulation with the residual that moves the cross-section, and the time-averaged cross-section is the object that absorbs the mixture—the results below hold against it, stated for the window-averaged regressor defined next.

Fix the cross-section as the within-window time average. For each location set  $\bar{\alpha}_j := \frac{1}{2}(\bar{\alpha}_j^1 + \bar{\alpha}_j^2)$ , the average composition across the two periods, and weight locations by mover incidence  $\mu_j$ , the share of mover endpoints at  $j$  counting origins and destinations (a symmetric weight that is well defined whether or not inflow equals outflow). Without stationarity, the period-specific regressor of (4) is no longer adapted to a single cross-section: in the large-location limit its composition term is  $\bar{\alpha}_{d(i)}^2 - \bar{\alpha}_{o(i)}^1$ , which evaluates the destination in period 2 and the origin in period 1—stationarity is precisely what collapses the two periods. Every result in this section is therefore stated for the *window-averaged* regressor

$$\delta_i^{(-i)} := \bar{Y}_{d(i)}^{(-i)} - \bar{Y}_{o(i)}^{(-i)}, \quad \bar{Y}_j^{(-i)} := \frac{1}{2}(\bar{Y}_{j,1}^{(-i)} + \bar{Y}_{j,2}^{(-i)}),$$

whose composition term in the large-location limit is  $\bar{\alpha}_{d(i)} - \bar{\alpha}_{o(i)}$ , with  $\bar{\alpha}_j$  the time-averaged composition. This is the Finkelstein et al. (2016) simple-year-average convention and the regressor implemented in Section VII of the paper; under Assumption 1 it coincides with (4) in the limit. The transition pair  $(\beta_\Delta, \lambda_\Delta)$ , the cross-sectional pair  $(\beta_M, \lambda_M)$ , and the dispersions  $V_{\psi_M}, V_{\psi_\Delta}$  are then sample functionals of  $(\psi_j, \bar{\alpha}_j, \bar{F})$  with  $\bar{\alpha}_j$  this single time-averaged composition.

**Theorem S2** (Recovery against the time-averaged cross-section). *Maintain the AKM model, exogenous mobility, and the large-location limit, but not Assumption 1, with  $\text{Var}_M(\Delta\psi_i) > 0$  and  $\text{Var}_M(\psi_j) > 0$ . Read the cross-section as the  $\mu$ -weighted distribution of  $(\psi_j, \bar{\alpha}_j)$  above, and let the mover regression and the transition moments use the window-averaged regressor  $\delta_i^{(-i)}$ . Then:*

- (i) **(Test.)** *The estimand identity  $\theta_\infty = f(\beta_\Delta, \lambda_\Delta)$ ,  $f(\beta, \lambda) = (1 + \beta)/((1 + \beta)^2 + \lambda)$ , holds, and the directional-isotropy restriction  $g(\vartheta) = (\beta_\Delta - \beta_M, Q_\Delta V_{\psi_M} - Q_M V_{\psi_\Delta})$  is a function of realized move-level and cross-sectional sample moments. The test  $H_0 : g(\vartheta) = 0$  of Section V is well defined and the inference of Section S4 is valid with no steady-state condition.*
- (ii) **(Recovery.)** *Directional isotropy,  $\beta_\Delta = \beta_M$  and  $\lambda_\Delta = \lambda_M$ , holds if and only if the mover design reproduces the cross-sectional sorting projection, and either implies  $\theta_\infty = \theta_{\text{iso}} = \frac{1}{2}(\theta_{\text{var}} + S_\psi^{\text{eq}})$  evaluated on this time-averaged cross-section, exactly as in Theorem 1 and Corollary 1.*
- (iii) **(Role of Assumption 1.)** *Dropping Assumption 1 leaves (i)–(ii) intact and only enlarges the set of flows consistent with a given cross-section—unbalanced, drifting configurations now qualify—so the range of  $\theta_\infty$  compatible with a fixed cross-section can only widen.*

*Proof.* (i) For a mover  $\Delta Y_i = \Delta\psi_i + \Delta\varepsilon_i$ , and the window-averaged regressor limits to  $\Delta\bar{Y}_i = \Delta\psi_i + \Delta\bar{\alpha}_i$  with  $\bar{\alpha}_j$  the time-averaged composition, so in the large-location limit  $\theta_\infty = \text{Cov}_M(\Delta\psi_i, \Delta\bar{Y}_i)/\text{Var}_M(\Delta\bar{Y}_i) = f(\beta_\Delta, \lambda_\Delta)$  by dividing numerator and denominator by  $\text{Var}_M(\Delta\psi_i)$ ; this is move-level algebra that never references  $\mathbb{E}_M[\Delta f_i]$ . Each component of  $g$  is a ratio of  $\mu$ -weighted sample (co)variances of  $(\psi_j, \bar{\alpha}_j)$  and of the move-level differences  $(\Delta\psi_i, \Delta\bar{\alpha}_i)$ , computed from the realized panel; the influence expansion of Proposition S2 differentiates these sample functionals directly, so Theorem S3 applies verbatim. (ii) The equivalence between  $\beta_\Delta = \beta_M, \lambda_\Delta = \lambda_M$  and directional isotropy, and the



implication  $\theta_\infty = \theta_{\text{iso}} = \frac{1}{2}(\theta_{\text{var}} + S_\psi^{\text{eq}})$ , are the ratio algebra of Theorem 1 and Corollary 1: they use only the definitions of the six moments and the projection  $f$ , all now formed on  $\bar{\alpha}_j$ , and at no point invoke  $\mathbb{E}_M[\Delta f_i] = 0$ . (iii) Mass balance is used exactly once in the paper, in the proof of Proposition 1, to drop the directed term when passing from  $F$  to  $\bar{F}$ ; no result in Sections III–V depends on it. The enlargement is Lemma S1 with its balance constraint removed: every cross-section and symmetric flow is induced by some assignment process, and lifting the constraint (S2) strictly enlarges the admissible set.  $\square$

The substance of (ii) is the two-sided link at the level of the *pair*  $(\beta, \lambda)$ , not the incidental scalar coincidence  $\theta_\infty = \theta_{\text{iso}}$  that  $f$  can produce off isotropy (cf. the remark after Theorem 1). What the design recovers is whether the realized routes trace the cross-section’s sorting geometry, and that question—and the test of it—stands free of any steady state. Stationarity buys only the clean undirected-flow picture of Proposition 1; the empirical conclusion that the routes traverse a different geometry than the cross-section they average over does not depend on it.

#### S4 Inference for the Isotropy Diagnostic

This section states the asymptotics and the multiplier bootstrap behind the joint isotropy test.

**Restriction map.** Collect the six cross-fitted second moments into

$$\vartheta = (a, b, c, d, e, f)' := (V_{\psi, M}, C_{\psi\alpha, M}, V_{\alpha, M}, V_{\psi, \Delta}, C_{\psi\alpha, \Delta}, V_{\alpha, \Delta})',$$

the first three the  $\mu$ -weighted cross-sectional moments of  $(\psi_j, \bar{\alpha}_j)$  and the last three the transition moments of  $(\Delta\psi_i, \Delta\bar{\alpha}_i)$ . The isotropy restrictions are the smooth map

$$g(\vartheta) = \begin{pmatrix} g^\beta \\ g^Q \end{pmatrix} = \begin{pmatrix} \frac{e}{d} - \frac{b}{a} \\ (f - \frac{e^2}{d})a - (c - \frac{b^2}{a})d \end{pmatrix}, \quad a, d > 0, \quad (\text{S3})$$

and  $g(\vartheta) = 0$  if and only if mobility is directionally isotropic (Theorem 1). The same  $\vartheta$  returns  $\theta_{\text{var}}, \theta_{\text{iso}}, S_\psi^{\text{eq}}$  through smooth maps, so every reported object is a function of one estimated moment vector.

**Cross-fitted estimation.** Split the sample in two. In split  $s \in \{1, 2\}$  estimate  $\hat{\psi}_j^s$  and  $\hat{Y}_j^s$ , and set  $\hat{\alpha}_j^s = \hat{Y}_j^s - \hat{\psi}_j^s$ . Each entry of  $\hat{\vartheta}$  is a symmetrized cross-split moment, e.g.  $\hat{V}_{\psi, M} = \frac{1}{2} \sum_j \mu_j (\hat{\psi}_j^1 - \hat{\psi}_j^2)(\hat{\psi}_j^2 - \hat{\psi}_j^1) + (1 \leftrightarrow 2)$ , and analogously over movers for the transition moments; the symmetrization is consequential for the cross moments such as  $\hat{C}_{\psi\alpha, M}$ , whose two split assignments differ. Independence across splits removes the own-sample bias that noisy  $\hat{\psi}_j$  would otherwise

inject into the quadratic moments.<sup>18</sup>

**Limit distribution.** The asymptotic frame is the one in which mover designs operate: fixed  $J$ , location sizes  $n_j \rightarrow \infty$ , independent workers, and connected split networks. In this regime asymptotic linearity is a *property* of the estimator, not an assumption. Every first-stage object is linear in outcomes:  $\widehat{\psi}^s$  solves the mover first-difference normal equations  $L_s \widehat{\psi}^s = A_s' \Delta Y^s$ , with  $L_s$  the split- $s$  mover-graph Laplacian and  $A_s$  its incidence matrix, and  $\widehat{Y}_j^s$  is a cell mean. Each entry of  $\widehat{\vartheta}$  is therefore an explicit quadratic form in worker-level data, and the influence of worker  $i$  on  $\widehat{\vartheta}$  is available in closed form,

$$\widehat{\varphi}_i = \underbrace{\widehat{\phi}_i^{\text{edge}}}_{\text{mover sampling}} + \underbrace{(W_{s(i)}[d(i)] - W_{s(i)}[o(i)]) \widehat{u}_i}_{\text{first stage: } \widehat{\psi}} + \underbrace{\sum_t \frac{1}{2} G_Y^{s(i)}[\cdot, j(i, t)] \frac{Y_{it} - \widehat{Y}_{j(i, t), t}^{s(i)}}{n_{j(i, t), t}^{s(i)}}}_{\text{first stage: cell means}}, \quad (\text{S4})$$

where  $\widehat{\phi}_i^{\text{edge}}$  is the mover's direct contribution to the six moment averages (for stayers both this term and the  $\widehat{\psi}$  term vanish, so stayers enter only through the cell means),  $\widehat{u}_i = \Delta Y_i - \Delta \widehat{\psi}_i$  is the mover's first-difference residual, the factor  $\frac{1}{2}$  appears because  $\widehat{Y}_j^s$  averages the two periods' cell means,  $G_\psi^s$  and  $G_Y^s$  are the closed-form gradients of  $\widehat{\vartheta}$  with respect to  $(\widehat{\psi}^s, \widehat{Y}^s)$ , and  $W_s := L_s^+ G_\psi^{s'}$  requires only six Laplacian solves per split. Because the  $\widehat{\psi}$ -term propagates through  $L_s^+$ , the dependence among movers sharing origins or destinations—and the correlation of  $\widehat{\psi}_j$  across connected locations generally—is carried exactly.

**Proposition S2** (Asymptotic linearity). *Fix  $J$  and let  $N \rightarrow \infty$  with  $n_j/N \rightarrow s_j > 0$  for all  $j$ , so that every location grows at rate  $N$ ; let workers be independent with uniformly bounded fourth moments of outcomes,  $\sup_{i,t} \mathbb{E}[Y_{it}^4] < \infty$ , and mover shares bounded away from zero, and let both split mover graphs be connected with probability approaching one. Then*

$$\widehat{\vartheta} - \vartheta = \sum_i \varphi_i + o_p(N^{-1/2}), \quad \sqrt{N}(\widehat{\vartheta} - \vartheta) \xrightarrow{d} \mathcal{N}(0, \Sigma_\vartheta), \quad \Sigma_\vartheta = \lim_N N \sum_i \mathbb{E}[\varphi_i \varphi_i'],$$

where  $\varphi_i$  is the population counterpart of (S4), and  $N \max_i \|\widehat{\varphi}_i - \varphi_i\| \xrightarrow{p} 0$ , so the estimated influences are uniformly accurate on the scale of a single contribution.

*Proof.* With  $J$  fixed,  $\widehat{\vartheta}$  is a smooth function of a finite-dimensional vector of sample means over independent workers: the cell sums and counts  $(\sum_i Y_{it} \mathbf{1}\{j(i, t) = j\}, n_{j,t})$  per split, the mover flow counts, and the mover cross-products entering the transition moments. Workers are independent but not identically distributed; the uniform fourth-moment bound delivers the Lyapunov condition,

<sup>18</sup>This two-way split corresponds to the  $(M = 2)$  case of the branch construction in Kline (2025). With fixed  $J$  and dense locations, a single naive split already produces independent estimates. This comes at the cost of discarding locations outside the intersection of the two connected split sets. Branching eliminates this loss in sparse, many-location networks, but that is not the regime studied in this paper.

so the multivariate CLT applies to the mean vector. On the event that both split graphs are connected—whose probability tends to one, so conditioning on it does not affect limits— $L_s$  has constant rank  $J - 1$  and  $L_s^+$  is a smooth function of the flow counts; the delta method then delivers the linear expansion. The gradient-weighted worker contributions coincide term by term with (S4), since differentiating through  $\widehat{\psi}^s = L_s^+ A_s' \Delta Y^s$  yields the  $W_s$  term and differentiating through the cell means yields the  $G_Y$  term. Consistency of  $\widehat{\varphi}_i$  follows from consistency of the gradients and residuals.  $\square$

With  $G = \partial g / \partial \vartheta'$ , the delta method gives  $\sqrt{N}(\widehat{g} - g) \xrightarrow{d} \mathcal{N}(0, \Omega)$ ,  $\Omega = G \Sigma_{\vartheta} G'$ . In the many-location regime ( $J \rightarrow \infty$  with moderate  $n_j$ ) the influence representation (S4) is unchanged but the CLT requires leverage conditions on the mover graph in the spirit of Kline et al. (2020); the fixed- $J$  regime is the one in which mover designs are applied and is maintained throughout.

**Multiplier bootstrap and test.** Draw  $\xi_i^{(b)} \stackrel{iid}{\sim} \mathcal{N}(0, 1)$  per worker and form  $\widehat{\vartheta}^{(b)} = \widehat{\vartheta} + \sum_i \xi_i^{(b)} \widehat{\varphi}_i$ , where  $\widehat{\varphi}_i$  is the raw, unscaled per-worker contribution to  $\widehat{\vartheta}$  from (S4)—each term is  $O_p(N^{-1})$ , so the perturbation is  $O_p(N^{-1/2})$ , the scale of  $\widehat{\vartheta} - \vartheta$  in Proposition S2. Set  $\widehat{g}^{(b)} = g(\widehat{\vartheta}^{(b)})$  and let  $\widehat{\Omega}_g$  be the sample covariance of  $\{\widehat{g}^{(b)} - \widehat{g}\}$ . No re-estimation is required: the draw is pushed through the fixed linear maps of (S4). The Wald statistic and its bootstrap analogue are

$$\mathcal{W} = \widehat{g}' \widehat{\Omega}_g^{-1} \widehat{g}, \quad \mathcal{W}^{(b)} = (\widehat{g}^{(b)} - \widehat{g})' \widehat{\Omega}_g^{-1} (\widehat{g}^{(b)} - \widehat{g}), \quad (\text{S5})$$

with bootstrap  $p$ -value  $B^{-1} \sum_b \mathbf{1}\{\mathcal{W}^{(b)} \geq \mathcal{W}\}$ . Evaluating the same draws through the cross-sectional maps yields percentile intervals for  $\theta_{\text{var}}, \theta_{\text{iso}}, S_{\psi}^{\text{eq}}$  and a one-sided bound for  $C$ ; a nonnegative bound supports the ordered reading of Corollary 1.

**Theorem S3 (Validity).** *Under the conditions of Proposition S2, with the  $\psi$ -dispersions  $a = V_{\psi, M}$  and  $d = V_{\psi, \Delta}$  bounded away from zero and  $\Omega$  positive definite,  $\mathcal{W} \xrightarrow{d} \chi_2^2$  under  $H_0 : g(\vartheta) = 0$ , and the multiplier law of  $\mathcal{W}^{(b)}$  consistently approximates the null law of  $\mathcal{W}$  conditional on the data.*

*Proof.* By Proposition S2 and the delta method,  $\sqrt{N} \widehat{g} \xrightarrow{d} \mathcal{N}(0, \Omega)$  under  $H_0$ . Conditionally on the data, the multiplier central limit theorem applies to  $\sum_i \xi_i^{(b)} \widehat{\varphi}_i$ : the outer-product sum  $\sum_i \widehat{\varphi}_i \widehat{\varphi}_i'$  is consistent for  $N^{-1} \Sigma_{\vartheta}$  by Proposition S2 and the law of large numbers, and the individual contributions are asymptotically negligible,  $\sqrt{N} \max_i \|\widehat{\varphi}_i\| \xrightarrow{p} 0$  under the fourth-moment bound. Hence  $\sqrt{N}(\widehat{\vartheta}^{(b)} - \widehat{\vartheta}) \xrightarrow{d} \mathcal{N}(0, \Sigma_{\vartheta})$  conditionally in probability; since  $g$  is smooth with  $a, d$  bounded away from zero, the mapped draws inherit the linearization,  $\widehat{\Omega}_g$  is consistent for  $N^{-1} \Omega$ , and the quadratic form converges to  $\chi_2^2$ . The same argument applied conditionally yields validity of the bootstrap  $p$ -value.  $\square$

**Remark S1 (Affine boundary).** When composition is one-dimensional,  $\lambda_M = \lambda_{\Delta} = 0$ , the parameter sits on the boundary of the parameter space:  $Q_M$  and  $Q_{\Delta}$  are variances and cannot fall below the null value, and the  $\sqrt{N}$ -scale variation of  $\widehat{g}_Q$  degenerates there—with  $\lambda = 0$  the cross-split

products are products of pure noise, of order  $N^{-1}$ —so the null distribution of  $\mathcal{W}$  is nonstandard rather than  $\chi^2_2$  (cf. Andrews, 2001), and the test is *conservative* (rejection 0.2–2.0% at the boundary in Table 1, versus 4.0–5.4% in the interior). The worker-level bootstrap keeps  $\widehat{\Omega}_g$  nonsingular in finite samples because first-stage noise moves  $\widehat{Q}_M$  and  $\widehat{Q}_\Delta$  even when  $\lambda = 0$ . Conservativeness at this point is innocuous: by Corollary 2(ii), the affine configuration is precisely where the SD-share reading is robust to any mobility pattern, so a missed rejection carries no interpretive cost. An exact-size boundary test (accounting for the nonnegativity of  $\lambda_M$  and  $\lambda_\Delta$  at the boundary, with chi-bar-squared critical values) is possible but unnecessary for the use made of the test here.

## S5 A Representation Lemma: The Reduced Form Is Without Loss of Generality

The analysis conditions on three reduced-form objects: the cross-section  $(\psi_j, \bar{\alpha}_j)$ , location sizes and mover shares  $(n_j, w_j)$ , and the symmetrized mover flow  $\bar{F}$ . The trivial direction is that any assignment process in the AKM class induces such a tuple. This section proves the converse: every admissible tuple is induced by some assignment process satisfying the paper’s maintained assumptions, so conditioning on the reduced form entails no loss of generality.

**Lemma S1** (Representation). *Let  $\mathcal{J}$  be a finite set of locations with attributes  $(\psi_j, \bar{\alpha}_j)_{j \in \mathcal{J}}$ , sizes  $n_j > 0$ , and mover shares  $w_j \in (0, 1)$ , and let  $\bar{F} = (\bar{F}_{jk})$  be any nonnegative symmetric flow with  $\bar{F}_{jj} = 0$  and  $\sum_k \bar{F}_{jk} = w_j n_j$  for all  $j$ . Then there exists an assignment process in the AKM class—a period-1 assignment rule, an origin-dependent destination rule, type distributions, and an error process—that satisfies exogenous mobility and Assumption 1 and induces exactly the cross-section  $(\psi_j, \bar{\alpha}_j)$ , the sizes and shares  $(n_j, w_j)$ , and the symmetrized flow  $\bar{F}$ .*

*Proof.* The proof is constructive. (i) *Population.* At each location  $j$ , place  $(1 - w_j)n_j$  stayers and  $w_j n_j$  movers in period 1. (ii) *Mover types.* Every mover, at every location, draws  $\alpha_i$  i.i.d. from a fixed distribution  $D$  with mean  $\bar{m}$  (say  $\bar{m} = 0$ ), independent of origin and destination. (iii) *Destinations.* A mover at  $j$  moves to  $k$  with probability  $K^2(k | j) = \bar{F}_{jk}/(w_j n_j)$ . The mass arriving at  $j$  is  $\sum_k \bar{F}_{kj} = w_j n_j$  by symmetry, so sizes and mover shares are constant across periods; arriving movers have type distribution  $D$ , equal to the distribution of those departing, so the mover balance condition of Assumption 1 holds exactly. (iv) *Stayer types.* Stayers at  $j$  draw  $\alpha_i$  from any distribution with mean  $(\bar{\alpha}_j - w_j \bar{m})/(1 - w_j)$ , so the location composition equals  $\bar{\alpha}_j$  in both periods. (v) *Outcomes.* Location effects are the given  $\psi_j$ , and  $\varepsilon_{it}$  is i.i.d. mean zero, independent of types and assignments, so exogenous mobility holds. The induced directed flow is  $F_{jk} = \bar{F}_{jk}$ , whose symmetrization is  $\bar{F}$ .  $\square$

Three remarks. First, the construction loads all compositional heterogeneity on stayers and makes movers homogeneous on average; this is enough because every estimand in the paper—the transition moments, the cross-sectional moments, and hence  $\theta_\infty$ , the pairs  $(\beta_\Delta, \lambda_\Delta)$  and  $(\beta_M, \lambda_M)$ , and the benchmark shares—depends on the assignment process only through  $(\psi, \bar{\alpha}, \bar{F})$ . Second, richer constructions keep movers sorted rather than homogeneous, and dispense with stayers entirely. Assigning a mover travelling between  $j$  and  $k$  a type centred on the symmetric edge

potential  $\varphi_j + \varphi_k$ , with  $\varphi$  solving  $(I + W)\varphi = \bar{\alpha}$  for  $W$  the row-normalised symmetrised flow, makes every location’s movers—leavers and arrivers alike—average its composition  $\bar{\alpha}_j$  in both periods, with no stayers required; the solve is exact whenever the flow is non-bipartite—bipartiteness being precisely the case in which  $-1$  is an eigenvalue of  $W$  and  $I + W$  fails to invert. This restriction is empirically vacuous: a symmetrized mover flow is bipartite only if the locations split into two classes with every move crossing between them and none within—equivalently, the mover network contains no odd cycle—a configuration ruled out by any three locations exchanging workers in a triangle, and thus absent from realized mover networks. A directional balancing,  $\sum_k \Phi_{jk} = \sum_k \Phi_{kj} = \deg_j \bar{\alpha}_j$  (the divergence-free condition of Section S2), drops the non-bipartiteness requirement but is itself feasible only on sufficiently connected supports, not on every flow with a cycle. The homogeneous construction of the lemma remains the universal witness precisely because stayers can absorb any composition; that is why the lemma itself needs no connectivity or parity condition at all. In each, a mover matches every location’s *mean* composition in both periods but, carrying one type across two firms, adds dispersion: movers share stayers’ location means, not their full distribution. The simulations of Section VI use this edge-potential economy. Third, by Proposition 1 any directed flow  $F$  with symmetrization  $\bar{F}$ —including heavily history-dependent ones—induces the same estimand, so the lemma’s symmetric choice of  $F$  is itself without loss of generality.

## S6 Descriptive Statistics for the Estimation Sample

Table 3 compares the estimation sample of Section VII—full-time male workers at firms above the 25th percentile of total worker-period observations—with the broader population of all full-time male stayers and one-time movers, over 2009–2016. The estimation sample is modestly older, slightly higher-paid, and sits at substantially larger firms, but is otherwise similar; cross-sector moves are a small share of moves in both.

Table 3: Descriptive Statistics: Estimation Sample versus Full Population

|                               | All stayers & movers<br>(full population) | Estimation sample<br>(large firms) |
|-------------------------------|-------------------------------------------|------------------------------------|
| Worker-year observations      | 10,621,264                                | 7,966,160                          |
| Age (years)                   | 46.7 (9.6)                                | 47.3 (9.4)                         |
| Annual gross earnings (euros) | 52,491 (50,060)                           | 53,998 (53,172)                    |
| Hourly wage (euros)           | 25.5 (24.5)                               | 26.2 (26.1)                        |
| Firm size (employees)         | 780 (811)                                 | 1,022 (798)                        |
| Mover (share of worker-years) | 0.30                                      | 0.28                               |
| Cross-sector mover (share)    | 0.06                                      | 0.05                               |

*Notes.* CBS linked employer–employee data, 2009–2016. The full population is all full-time male stayers and one-time movers; the estimation sample additionally restricts to firms above the 25th percentile of total worker-period observations (Section VII). All entries are worker-year means, with standard deviations in parentheses for continuous variables. Hourly wage is gross annual earnings spread over a 52-week, 40-hour year. “Mover” is the worker-year share of one-time movers (the worker-level one-time-mover share in the estimation sample is 17.0 percent), and “cross-sector mover” the share whose move changes 1-digit sector. Firm size is the employer’s total headcount from register size-intervals.

## S7 Robustness: Unadjusted Log Hourly Wage

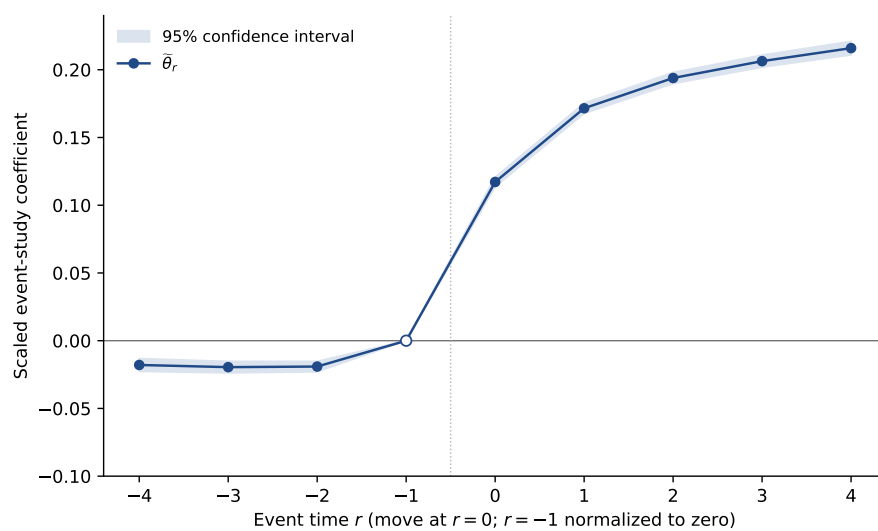
The paper's outcome is the log gross hourly wage residualized on age and calendar-year indicators in a preliminary step. This section repeats the full analysis on the *unadjusted* log hourly wage, on the identical sample. Every conclusion is unchanged: the affine benchmark is rejected ( $\hat{\lambda}_M = 2.52$ , CI [2.31, 2.76]), both isotropy restrictions reject, and the joint Wald is 118.2 ( $\chi^2_2 p = 2.2 \times 10^{-26}$ ). Magnitudes shift by roughly ten percent relative to the adjusted outcome; no sign or significance changes. Figure 2 shows the event study: pre-event coefficients are slightly larger than in the adjusted specification (up to 0.020 in absolute value versus 0.015), consistent with the age and year indicators absorbing life-cycle and calendar drift, which motivates the adjusted outcome as the main specification.

Table 4: Mover-Design Diagnostics, Unadjusted Log Hourly Wage

|                                                                         | Estimate | SE    | 95% CI           |
|-------------------------------------------------------------------------|----------|-------|------------------|
| <i>Panel A. Mover-design coefficient and cross-sectional benchmarks</i> |          |       |                  |
| Mover regression $\hat{\theta}$                                         | 0.238    | 0.002 | [0.234, 0.242]   |
| Transition projection $f(\hat{\beta}_\Delta, \hat{\lambda}_\Delta)$     | 0.224    | 0.004 | [0.215, 0.232]   |
| $\theta_{\text{iso}}$ (midpoint)                                        | 0.291    | 0.007 | [0.277, 0.305]   |
| $\theta_{\text{var}}$                                                   | 0.275    | 0.015 | [0.246, 0.305]   |
| $\theta_{\text{sd}}$                                                    | 0.524    | 0.014 | [0.496, 0.552]   |
| $S_\psi^{\text{eq}}$                                                    | 0.307    | 0.015 | [0.277, 0.333]   |
| <i>Panel B. Sorting geometry: cross-section versus transitions</i>      |          |       |                  |
| Cross-sectional slope $\hat{\beta}_M$                                   | 0.058    | 0.049 | [-0.029, 0.163]  |
| Transition slope $\hat{\beta}_\Delta$                                   | -0.087   | 0.025 | [-0.133, -0.036] |
| Cross-sectional residual intensity $\hat{\lambda}_M$                    | 2.521    | 0.114 | [2.306, 2.758]   |
| Transition residual intensity $\hat{\lambda}_\Delta$                    | 3.249    | 0.105 | [3.047, 3.473]   |
| <i>Panel C. Isotropy test</i>                                           |          |       |                  |
| Slope restriction $\hat{g}_\beta$                                       | -0.144   | 0.044 | [-0.234, -0.064] |
| Residual restriction $\hat{g}_Q (\times 10^4)$                          | 1.027    | 0.181 | [0.676, 1.393]   |
| Joint Wald $\mathcal{W}$ ( $\chi^2_2 p = 2.2 \times 10^{-26}$ )         | 118.190  | —     | —                |

*Notes.* Identical sample, implementation, and inference as Table 2 of the paper; the only change is that the outcome is the raw log gross hourly wage. Bootstrap  $p$ -value of the joint Wald below 0.001.

Figure 2: Event Study, Unadjusted Log Hourly Wage



Notes. As in Figure 1 of the paper, with the unadjusted log hourly wage as the outcome.